

Title: Design and Implementation of a Campus Network Using Cisco Packet Tracer.

Objectives:

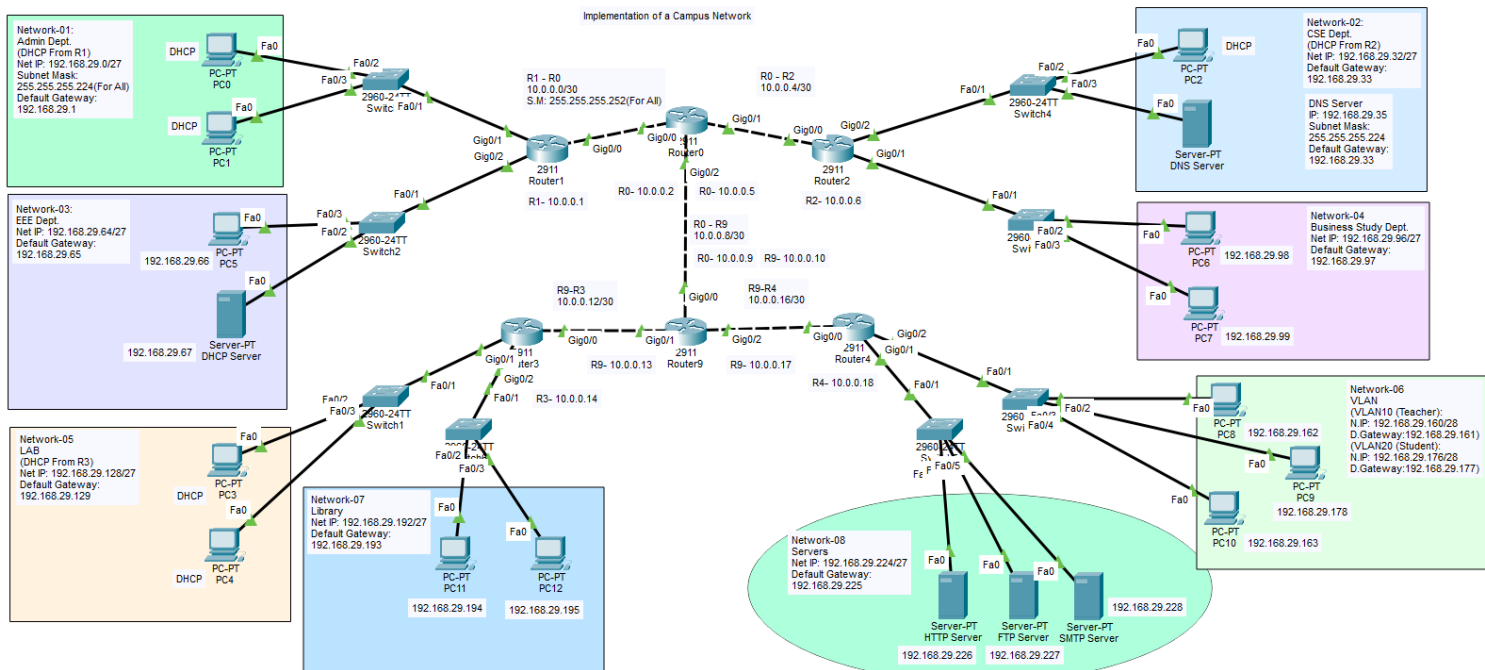
By end of this project we will be able to know about:

1. To design a campus network using Cisco Packet Tracer.
2. To connect different departments through routers and switches.
3. To divide the network into smaller subnetworks.
4. To implement inter-router communication through routing. And how OSPF Dynamic routing protocol configured to all router.
5. To implement VLAN for better network management.
6. To configure DHCP for automatic IP address assignment.
7. To configure DNS for name resolution.
8. To set up HTTP, FTP, and SMTP servers.
9. To test communication between all departments.
10. To understand real-life networking concepts through practical implementation.

Introduction:

In today's world, computer networks play an important role in communication and resource sharing. Educational institutions, offices, and organizations use campus networks to connect different departments and users. A campus network allows users to share information, access services, and communicate efficiently within the same environment. This project focuses on the implementation of a campus network using Cisco Packet Tracer. The network consists of several departments including the Administration Department, CSE Department, EEE Department, Business Studies Department, Laboratory, Library, and Server Room. Each department is connected through routers and switches to create a complete network infrastructure. To improve network performance and organization, subnetting and VLAN technologies were used. DHCP was configured to assign IP addresses automatically, while DNS was used for name resolution. Different servers such as HTTP, FTP, and SMTP servers were also implemented to provide web, file transfer, and email services. The main purpose of this project is to gain practical knowledge of network design, configuration, and management while simulating a real-world campus network environment.

Theory:



1/Campus Network:

A campus network is a network infrastructure that connects different departments, offices, laboratories, libraries, and servers within the same institution or organization. It allows users to communicate, share resources, and access network services efficiently. In this project, a complete campus network was implemented using Cisco Packet Tracer to simulate a real-world educational environment. The network includes multiple departments such as the Administration Department, CSE Department, EEE Department, Business Studies Department, Laboratory, Library, and Server Room. These departments are interconnected through routers and switches to ensure seamless communication.

Advantages of Campus Network:

- I. Centralized network management.
- II. Easy resource sharing among departments.
- III. Faster communication between users.
- IV. Improved network security.
- V. Easy maintenance and scalability.

Configuration Tasks Performed:

- I. Designed the physical and logical network topology.
- II. Connected routers, switches, PCs, and servers.
- III. Assigned IP addresses to all network devices.
- IV. Configured default gateways for communication between networks.
- V. Tested network connectivity using ping commands.

2/Subnetting:

Subnetting is the process of dividing a large network into smaller and more manageable subnetworks. It helps reduce broadcast traffic, improve network performance, and utilize IP addresses efficiently. For this project, the main network 192.168.29.0/24 was divided into several subnetworks based on departmental requirements.

Subnet Allocation Table:

Department	Network Address
Admin Department	192.168.29.0/27
CSE Department	192.168.29.32/27
EEE Department	192.168.29.64/27
Business Studies Department	192.168.29.96/27
Laboratory	192.168.29.128/27
VLAN 10	192.168.29.160/28
VLAN 20	192.168.29.176/28
Library	192.168.29.192/27
Server Network	192.168.29.224/27

Benefits of Subnetting:

- I. Efficient use of IP addresses.
- II. Reduced broadcast domains.
- III. Better traffic management.
- IV. Improved security.
- V. Easier troubleshooting.

Configuration Tasks Performed:

- I. Calculated subnet masks and network ranges.
- II. Assigned network addresses and default gateways.
- III. Configured router interfaces with appropriate IP addresses.
- IV. Verified subnet connectivity using ping tests.

3/WAN (Wide Area Network):

WAN (Wide Area Network) is a type of network that connects multiple LANs over a large geographical area. Unlike a LAN, which covers a single building or department, a WAN can connect different networks located in separate locations.

Why WAN Was Used in This Project:

- I. To connect multiple routers together.
- II. To enable communication between different subnetworks.
- III. To provide access to centralized servers.
- IV. To allow routing between departments.
- V. To simulate a real-world enterprise or campus network environment.

Router-To-Router link:

Link	Network
R1-R0	10.0.0.0/30
R0-R2	10.0.0.4/30
R0-R9	10.0.0.8/30
R9-R3	10.0.0.12/30
R9-R4	10.0.0.16/30

4/VLAN (Virtual Local Area Network):

VLAN stands for Virtual Local Area Network. It is used to logically divide users into different groups even when they are connected to the same physical switch. In this project:

- i. VLAN 10 was assigned to Teachers.
- ii. VLAN 20 was assigned to Students.

Advantages of VLAN:

- i. Improved network security.
- ii. Reduced broadcast traffic.
- iii. Better network organization.

- iv. Easier administration and management.

Configuration Tasks Performed:

- i. Created VLAN 10 and VLAN 20 on the switch.
- ii. Assigned switch ports to their respective VLANs.
- iii. Configured trunk links between switches and routers.
- iv. Implemented Inter-VLAN Routing to allow communication between VLANs.
- v. Verified VLAN connectivity through testing.

Configuration Process:

1. Switch:

```
enable  
conf t
```

```
vlan 10  
name STUDENT
```

```
vlan 20  
name TEACHER
```

2. VLAN 10 Ports:

```
interface fa0/2  
switchport mode access  
switchport access vlan 10
```

```
interface fa0/4  
switchport mode access  
switchport access vlan 10
```

3. VLAN 20 Ports:

```
interface fa0/3  
switchport mode access  
switchport access vlan 20
```

4. Trunk Port:

```
interface fa0/1  
switchport mode trunk
```

Using VLANs helps reduce unnecessary traffic and increases security because users from different VLANs are separated logically.

5/ Routing:

Routing is the process of forwarding data packets between different networks. Routers determine the best path for data transmission and ensure successful communication between departments.

Since multiple subnetworks exist in this project, routing was essential for connecting all departmental networks.

Types of Routing:

1. **Static Routing:** Static Routing is a routing method where the network administrator manually configures routes on the router.
2. **Dynamic Routing:** Dynamic Routing is a routing method where routers automatically exchange routing information with each other. Routers use routing protocols to learn the best path for forwarding packets.

Common dynamic routing protocols include:

1. RIP (Routing Information Protocol)
2. OSPF (Open Shortest Path First)
3. EIGRP (Enhanced Interior Gateway Routing Protocol)

3. **Default Routing:** Default Routing is a routing method where all unknown network traffic is sent to a specific router called the default gateway. It is used when the router does not know the exact destination network.

This project mainly used routing configurations to establish communication among all subnetworks.

Importance of Routing:

- i. Connects different networks.
- ii. Enables inter-department communication.
- iii. Supports network expansion.
- iv. Improves network efficiency.

Configuration Tasks Performed:

- i. Assigned IP addresses to router interfaces.
- ii. Enabled router interfaces using the no shutdown command.
- iii. Configured routing tables.
- iv. Tested communication between different networks.

Configuration Process:

1. Router 1:

```
enable
configure terminal

hostname Router1

interface g0/1
ip address 192.168.29.1 255.255.255.224
no shutdown

interface g0/2
ip address 192.168.29.65 255.255.255.224
no shutdown

interface g0/0
ip address 10.0.0.1 255.255.255.252
no shutdown

router ospf 1
network 192.168.29.0 0.0.0.31 area 0
network 192.168.29.64 0.0.0.31 area 0
network 10.0.0.0 0.0.0.3 area 0

end
wr
```

2. Router 0:

```
enable
configure terminal

hostname Router0

interface g0/0
ip address 10.0.0.2 255.255.255.252
no shutdown

interface g0/1
ip address 10.0.0.5 255.255.255.252
no shutdown

interface g0/2
ip address 10.0.0.9 255.255.255.252
no shutdown
```

```
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
```

```
end
wr
```

3. Router 2:

```
enable
configure terminal
```

```
hostname Router2
```

```
interface g0/2
ip address 192.168.29.33 255.255.255.224
no shutdown
```

```
interface g0/1
ip address 192.168.29.97 255.255.255.224
no shutdown
```

```
interface g0/0
ip address 10.0.0.6 255.255.255.252
no shutdown
```

```
router ospf 1
network 192.168.29.32 0.0.0.31 area 0
network 192.168.29.96 0.0.0.31 area 0
network 10.0.0.4 0.0.0.3 area 0
```

```
end
wr
```

4. Router 3:

```
enable
configure terminal
```

```
hostname Router3
```

```
interface g0/1
ip address 192.168.29.129 255.255.255.224
no shutdown
```

```
interface g0/2
ip address 192.168.29.193 255.255.255.224
no shutdown
```

```
interface g0/0
ip address 10.0.0.14 255.255.255.252
no shutdown
```

```
router ospf 1
network 192.168.29.128 0.0.0.31 area 0
network 192.168.29.192 0.0.0.31 area 0
network 10.0.0.12 0.0.0.3 area 0
```

```
end
wr
```

5. Router 9:

```
enable
configure terminal
```

```
hostname Router9
```

```
interface g0/0
ip address 10.0.0.10 255.255.255.252
no shutdown
```

```
interface g0/1
ip address 10.0.0.13 255.255.255.252
no shutdown
```

```
interface g0/2
ip address 10.0.0.17 255.255.255.252
no shutdown
```

```
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
```

```
end
wr
```

6. Router 4:

```
enable
configure terminal
```

```
hostname Router4
```

```
interface g0/0
ip address 10.0.0.18 255.255.255.252
no shutdown
```

```
interface g0/2
ip address 192.168.29.225 255.255.255.224
no shutdown
```

Router-on-a-Stick:

```
interface g0/1
no shutdown
```

```
interface g0/1.10
encapsulation dot1Q 10
ip address 192.168.29.161 255.255.255.240
```

```
interface g0/1.20
encapsulation dot1Q 20
ip address 192.168.29.177 255.255.255.240
```

OSPF:

```
router ospf 1
```

```
network 10.0.0.16 0.0.0.3 area 0
network 192.168.29.160 0.0.0.15 area 0
network 192.168.29.176 0.0.0.15 area 0
network 192.168.29.224 0.0.0.31 area 0
```

```
end
wr
```

6/ DHCP (Dynamic Host Configuration Protocol):

DHCP stands for Dynamic Host Configuration Protocol. It automatically assigns IP addresses, subnet masks, default gateways, and DNS information to client devices..

Configuration Process:

1. Network: 01 (Admin)

```
ip dhcp excluded-address 192.168.29.1
```

```
ip dhcp pool ADMIN
```

```
network 192.168.29.0 255.255.255.224
```

```
default-router 192.168.29.1
```

```
dns-server 192.168.29.35
```

2. Network: 02 (CSE)

```
ip dhcp excluded-address 192.168.29.33
```

```
ip dhcp pool CSE
```

```
network 192.168.29.32 255.255.255.224
```

```
default-router 192.168.29.33
```

```
dns-server 192.168.29.35
```

3. Network: 05 (LAB)

```
ip dhcp excluded-address 192.168.29.129
```

```
ip dhcp pool LAB
```

```
network 192.168.29.128 255.255.255.224
```

```
default-router 192.168.29.129
```

```
dns-server 192.168.29.35
```

7/ DNS (Domain Name System):

DNS (Domain Name System) converts domain names into IP addresses so that users can access services easily.

Server:

IP = 192.168.29.35

Mask = 255.255.255.224

Gateway = 192.168.29.35

- ❖ Click Services then go to DNS server and select ON.

Records:

www.campus.com and give server IP here 192.168.29.226 .

8/ HTTP Server (Hypertext Transfer Protocol):

The HTTP server provides web services and allows users to access web pages through a browser.

Server:

IP = 192.168.29.226

Gateway = 192.168.29.225

- ❖ Click Services then go to HTTP server and select ON.
- ❖ Then select index.html and edit it by writing

```
<html>
```

```
Welcome to Campus Network!
```

```
</html>
```

- ❖ Now save this. And check any PC web browser as it working or not.

9/ FTP Server (File Transfer Protocol):

The FTP server allows users to upload and download files across the network.

Server:

IP = 192.168.29.227

Gateway = 192.168.29.225

❖ Click Services then go to FTP server and select ON.Record:

Campus Network – Pas: 123

10/ SMTP Server (Simple Mail Transfer Protocol):

The SMTP server is responsible for sending and managing email communication between users.

Server:

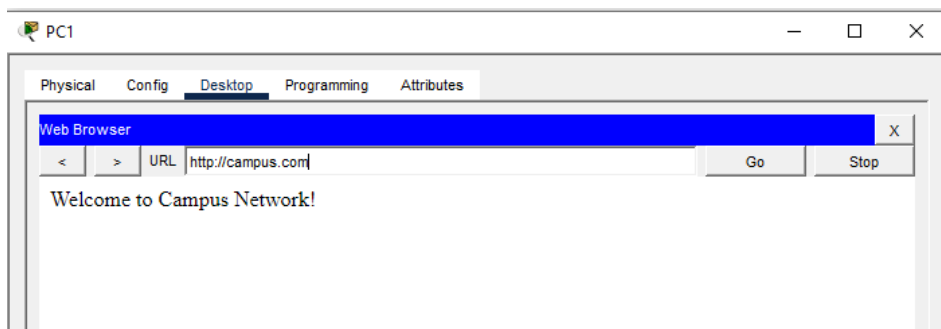
IP = 192.168.29.228

Gateway = 192.168.29.225

❖ Click Services then go to EMAIL server and select ON.

Domain: **google.com**

User: **User1 & User2**





Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Request timed out.
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 10ms, Average = 10ms

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125
Reply from 192.168.29.35: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time=10ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time=18ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
```



Physical Config Desktop Programming Attributes

Command Prompt

```
ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

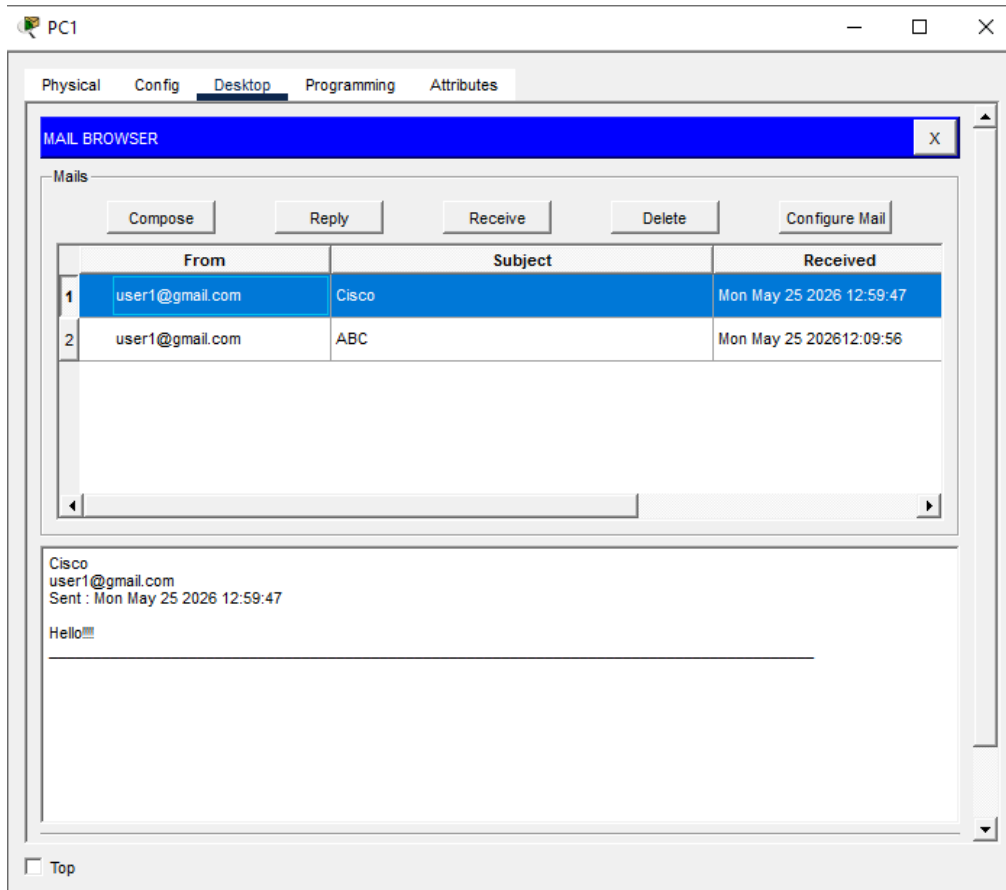
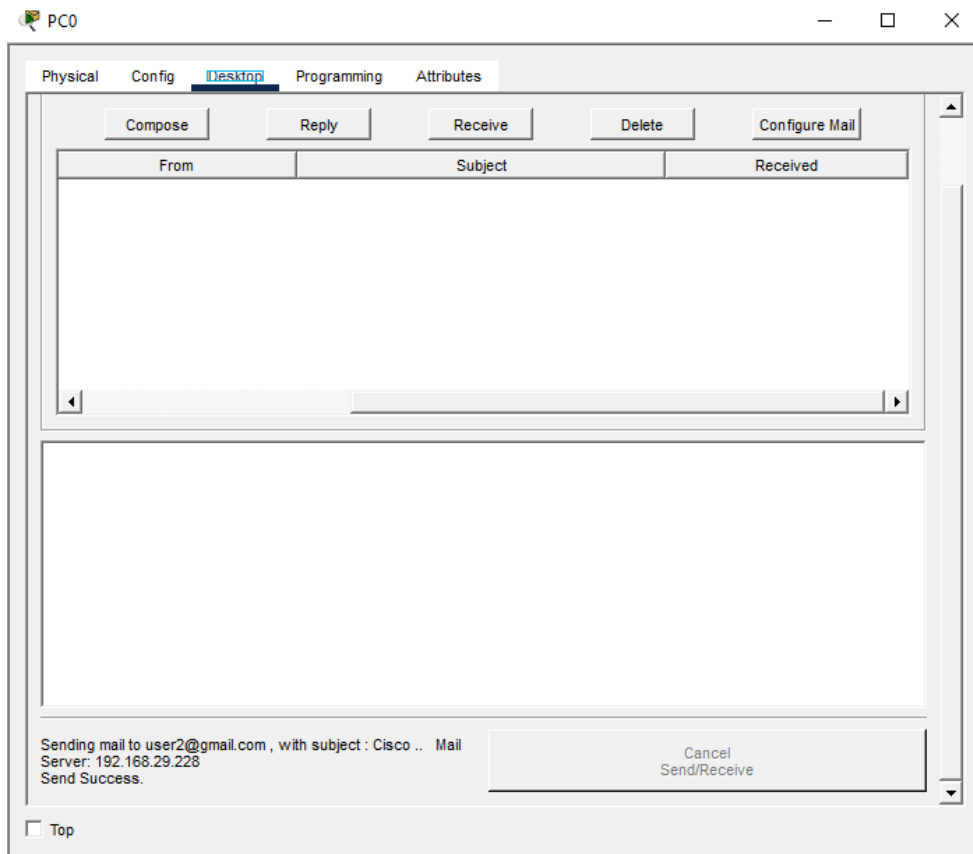
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text                               28
 1 : asa842-k8.bin                             5571584
 2 : asa923-k8.bin                             30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin    33591768
 4 : c1841-ipbase-mz.123-14.T7.bin             13832032
 5 : c1841-ipbasek9-mz.124-12.bin              16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin     33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin    33591768
 8 : c2600-i-mz.122-28.bin                     5571584
 9 : c2600-ipbasek9-mz.124-8.bin                13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin   50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin    33591768
12 : c2800nm-ipbase-mz.123-14.T7.bin            5571584
13 : c2800nm-ipbasek9-mz.124-8.bin              15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin     33591768
15 : c2950-i6q412-mz.121-22.EA4.bin            3058048
16 : c2950-i6q412-mz.121-22.EA8.bin            3117390
17 : c2960-lanbase-mz.122-25.FX.bin             4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin           4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin           4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin    8662192
21 : c3560-advipservicesk9-mz.122-46.SE1.bin    10713279
22 : c800-universalk9-mz.SPA.152-4.M4.bin        33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin       83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin      505832849
25 : cgr1000-universalk9-mz.SPA.154-2.CG         159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG         184530138
27 : ie9k_iosxe.17.09.04.SPA.bin                596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin    160968869
29 : ir800-universalk9-mz.SPA.155-3.M            61750062
30 : ir800-universalk9-mz.SPA.156-3.M            63753767
31 : ir800_yocto-1.7.2.tar                       2877440
32 : ir800_yocto-1.7.2.python-2.7.3.tar         6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin                     5571584
35 : pt3000-i6q412-mz.121-22.EA4.bin            3117390

ftp>|
```





PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125
Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time=11ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time=10ms TTL=124
Reply from 192.168.29.226: bytes=32 time=3ms TTL=124

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 6ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time=16ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 6ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=18ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 7ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
```



PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 6ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=18ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 7ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text                               28
 1 : asa842-k8.bin                             5571584
 2 : asa923-k8.bin                             30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin   33591768
 4 : c1841-ibase-mz.123-14.T7.bin             13832032
 5 : c1841-ibasek9-mz.124-12.bin              16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin   33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin   33591768
 8 : c2600-i-mz.122-28.bin                    5571584
 9 : c2600-ibasek9-mz.124-8.bin               13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin   33591768
12 : c2800nm-ibase-mz.123-14.T7.bin           5571584
13 : c2800nm-ibasek9-mz.124-8.bin             15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin   33591768
15 : c2950-i6q4l2-mz.121-22.EA4.bin          3058048
16 : c2950-i6q4l2-mz.121-22.EA8.bin          3117390
17 : c2960-lanbase-mz.122-25.FX.bin           4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin         4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin         4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin  8662192
21 : c3560-advipservicesk9-mz.122-46.SE1.bin  10713279
22 : c800-universalk9-mz.SPA.152-4.M4.bin      33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin     83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin   505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG      159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG      184530138
27 : ie9k_iosxe.17.09.04.SPA.bin             596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
29 : ir800-universalk9-mz.SPA.155-3.M         61750062
30 : ir800-universalk9-mz.SPA.156-3.M         63753767
31 : ir800_yocto-1.7.2.tar                    2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar      6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin                   5571584
35 : pt3000-i6q4l2-mz.121-22.EA4.bin          3117390

ftp>
```

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time<1ms TTL=125
Reply from 192.168.29.35: bytes=32 time=10ms TTL=125
Reply from 192.168.29.35: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time=10ms TTL=124
Reply from 192.168.29.226: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time=1ms TTL=124
Reply from 192.168.29.227: bytes=32 time=17ms TTL=124
Reply from 192.168.29.227: bytes=32 time=11ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 7ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124
Reply from 192.168.29.228: bytes=32 time=11ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

```

Command Prompt

```

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 7ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

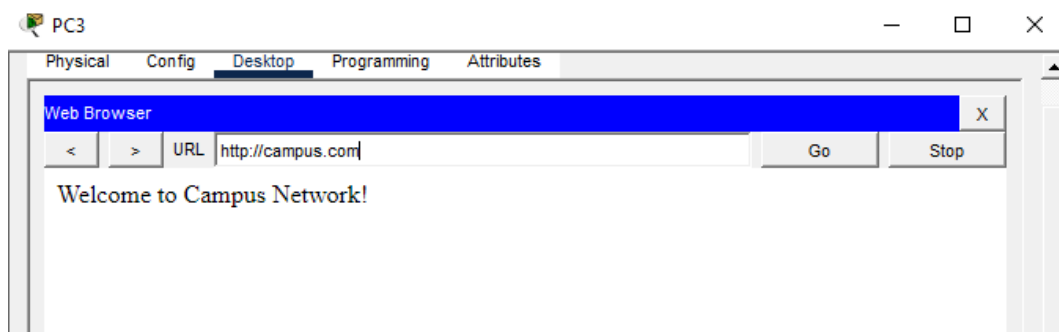
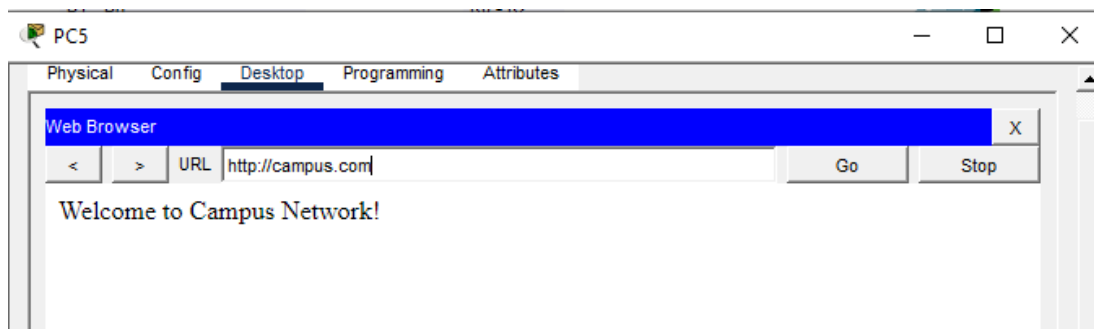
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124
Reply from 192.168.29.228: bytes=32 time=11ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text                               28
 1 : asa842-k8.bin                             5571584
 2 : asa923-k8.bin                             30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin    33591768
 4 : c1841-ipbase-mz.123-14.T7.bin             13832032
 5 : c1841-ipbasek9-mz.124-12.bin             16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin    33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin    33591768
 8 : c2600-i-mz.122-28.bin                     5571584
 9 : c2600-ipbasek9-mz.124-8.bin               13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin    33591768
12 : c2800nm-ipbase-mz.123-14.T7.bin           5571584
13 : c2800nm-ipbasek9-mz.124-8.bin            15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin    33591768
15 : c2950-i6q4l2-mz.121-22.EA4.bin           3058048
16 : c2950-i6q4l2-mz.121-22.EA8.bin           3117390
17 : c2960-lanbase-mz.122-25.FX.bin            4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin          4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin          4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin   8662192
21 : c3560-advipservicesk9-mz.122-46.SE1.bin   10713279
22 : c800-universalk9-mz.SPA.152-4.M4.bin       33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin     83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin    505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG       159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG       184530138
27 : ie9k_iosxe.17.09.04.SPA.bin              596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
29 : ir800-universalk9-mz.SPA.155-3.M          61750062
30 : ir800-universalk9-mz.SPA.156-3.M          63753767
31 : ir800_yocto-1.7.2.tar                     2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar       6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin                    5571584
35 : pt3000-i6q4l2-mz.121-22.EA4.bin          3117390
ftp>

```





Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Request timed out.
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time=10ms TTL=124
Reply from 192.168.29.35: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 6ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time=10ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=10ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir
```



Physical Config Desktop Programming Attributes

Command Prompt

```
Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=10ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text                               28
 1 : asa842-k8.bin                             5571584
 2 : asa923-k8.bin                             30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin    33591768
 4 : c1841-ibasek9-mz.123-14.T7.bin           13832032
 5 : c1841-ibasek9-mz.124-12.bin               16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin    33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin    33591768
 8 : c2600-i-mz.122-28.bin                     5571584
 9 : c2600-ibasek9-mz.124-8.bin                13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4a.bin   33591768
12 : c2800nm-ibasek9-mz.123-14.T7.bin          5571584
13 : c2800nm-ibasek9-mz.124-8.bin              15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin    33591768
15 : c2950-i6q412-mz.121-22.EA4.bin           3058048
16 : c2950-i6q412-mz.121-22.EA8.bin           3117390
17 : c2960-lanbase-mz.122-25.FX.bin            4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin          4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin          4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin   8662192
21 : c3560-advipservicesk9-mz.122-46.SE1.bin   10713279
22 : c800-universalk9-mz.SPA.152-4.M4a.bin     33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin     83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin    505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG       159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG       184530138
27 : ie9k_iosxe.17.09.04.SPA.bin              596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
29 : ir800-universalk9-mz.SPA.155-3.M          61750062
30 : ir800-universalk9-mz.SPA.156-3.M          63753767
31 : ir800_yocto-1.7.2.tar                     2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar       6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin                    5571584
35 : pt3000-i6q412-mz.121-22.EA4.bin          3117390
ftp>
```

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
 Reply from 192.168.29.35: bytes=32 time=10ms TTL=124
 Reply from 192.168.29.35: bytes=32 time=11ms TTL=124
 Reply from 192.168.29.35: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.35:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.226: bytes=32 time=14ms TTL=125
 Reply from 192.168.29.226: bytes=32 time=1ms TTL=125
 Reply from 192.168.29.226: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.226:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.227: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.227:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.29.228:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password: ***

230- Logged in

(passive mode On)

Command Prompt

Ping statistics for 192.168.29.227:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
 Reply from 192.168.29.228: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.29.228:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password: ***

230- Logged in

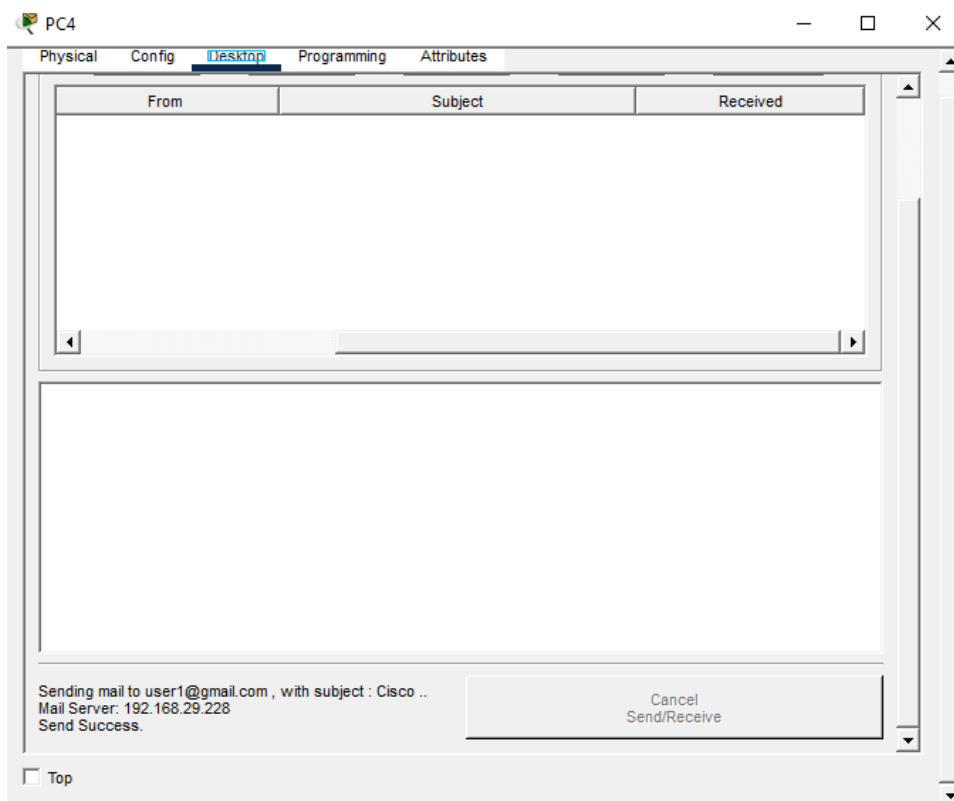
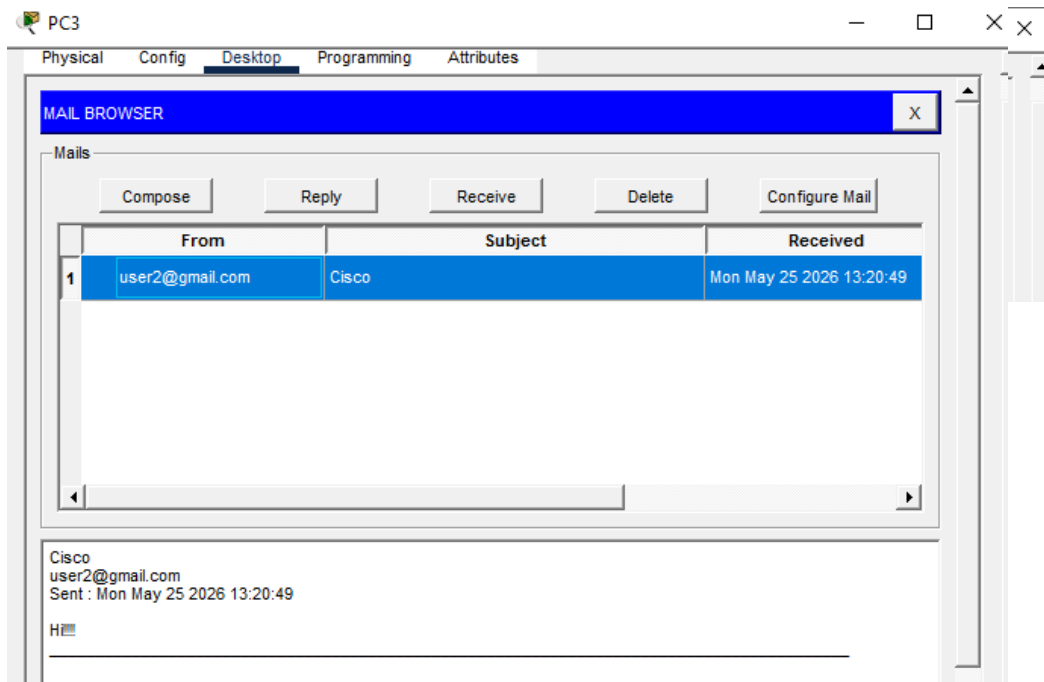
(passive mode On)

ftp>dir

Listing /ftp directory from 192.168.29.227:

0	: admin.text	28
1	: asa842-k8.bin	5571584
2	: asa923-k8.bin	30468096
3	: c1841-advipservicesk9-mz.124-15.T1.bin	33591768
4	: c1841-ipbase-mz.123-14.T7.bin	13832032
5	: c1841-ipbasek9-mz.124-12.bin	16599160
6	: c1900-universalk9-mz.SPA.155-3.M4a.bin	33591768
7	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
8	: c2600-i-mz.122-28.bin	5571584
9	: c2600-ipbasek9-mz.124-8.bin	13169700
10	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
11	: c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768
12	: c2800nm-ipbase-mz.123-14.T7.bin	5571584
13	: c2800nm-ipbasek9-mz.124-8.bin	15522644
14	: c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
15	: c2950-i6q412-mz.121-22.EA4.bin	3058048
16	: c2950-i6q412-mz.121-22.EA8.bin	3117390
17	: c2960-lanbase-mz.122-25.FX.bin	4414921
18	: c2960-lanbase-mz.122-25.SE1.bin	4670455
19	: c2960-lanbasek9-mz.150-2.SE4.bin	4670455
20	: c3560-advipservicesk9-mz.122-37.SE1.bin	8662192
21	: c3560-advipservicesk9-mz.122-46.SE1.bin	10713279
22	: c800-universalk9-mz.SPA.152-4.M4.bin	33591768
23	: c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
24	: cat3k_caa-universalk9.16.03.02.SPA.bin	505532849
25	: cgr1000-universalk9-mz.SPA.154-2.CG	159487552
26	: cgr1000-universalk9-mz.SPA.156-3.CG	184530138
27	: ie9k_iosxe.17.09.04.SPA.bin	596133776
28	: ir800-universalk9-bundle.SPA.156-3.M.bin	160968869
29	: ir800-universalk9-mz.SPA.155-3.M	61750062
30	: ir800-universalk9-mz.SPA.156-3.M	63753767
31	: ir800_yocto-1.7.2.tar	2877440
32	: ir800_yocto-1.7.2_python-2.7.3.tar	6912000
33	: ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg	685645824
34	: pt1000-i-mz.122-28.bin	5571584
35	: pt3000-i6q412-mz.121-22.EA4.bin	3117390

ftp>



Physical Config Desktop Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to FT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

```

Physical Config Desktop Programming Attributes

Command Prompt

```

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

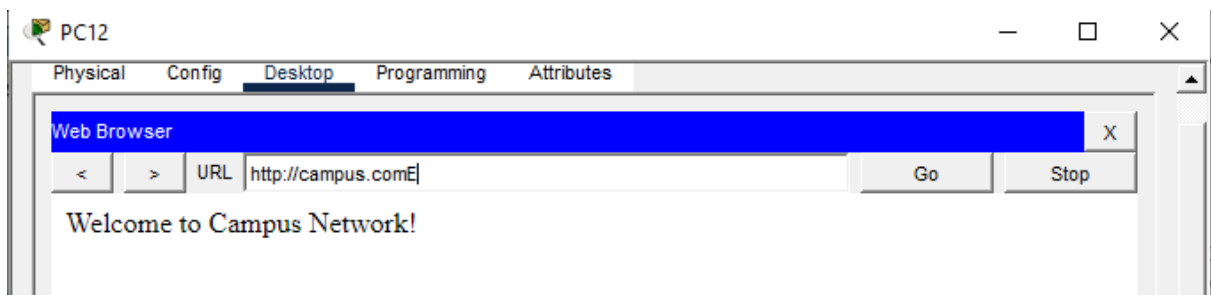
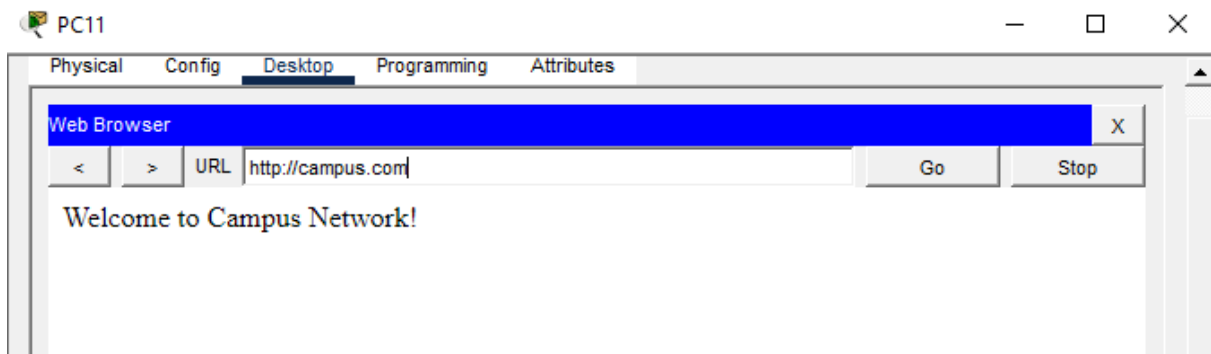
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to FT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0  : admin.text                               28
 1  : asa842-k8.bin                           5571584
 2  : asa923-k8.bin                           30468096
 3  : c1841-advipservicesk9-mz.124-15.T1.bin  33591768
 4  : c1841-ibasek9-mz.123-14.T7.bin          13832032
 5  : c1841-ibasek9-mz.124-12.bin             16599160
 6  : c1900-universalk9-mz.SPA.155-3.M4a.bin  33591768
 7  : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
 8  : c2600-i-mz.122-28.bin                   5571584
 9  : c2600-ibasek9-mz.124-8.bin              13169700
10  : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
11  : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
12  : c2800nm-ibasek9-mz.123-14.T7.bin        5571584
13  : c2800nm-ibasek9-mz.124-8.bin            15522644
14  : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
15  : c2950-i6q412-mz.121-22.EA4.bin         3058048
16  : c2950-i6q412-mz.121-22.EA8.bin         3117390
17  : c2960-lanbase-mz.122-25.FX.bin          4414921
18  : c2960-lanbase-mz.122-25.SEE1.bin        4670455
19  : c2960-lanbasek9-mz.150-2.SE4.bin        4670455
20  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
21  : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
22  : c800-universalk9-mz.SPA.152-4.M4a.bin    33591768
23  : c800-universalk9-mz.SPA.154-3.M6a.bin    83029236
24  : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
25  : cgr1000-universalk9-mz.SPA.154-2.CG     159487552
26  : cgr1000-universalk9-mz.SPA.156-3.CG     184530138
27  : ie9k_iosxe.17.09.04.SPA.bin            596133776
28  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
29  : ir800-universalk9-mz.SPA.155-3.M        61750062
30  : ir800-universalk9-mz.SPA.156-3.M        63753767
31  : ir800_yocto-1.7.2.tar                   2877440
32  : ir800_yocto-1.7.2_python-2.7.3.tar     6912000
33  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34  : pt1000-i-mz.122-28.bin                  5571584
35  : pt3000-i6q412-mz.121-22.EA4.bin        3117390
ftp>

```



Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35
```

```
Pinging 192.168.29.35 with 32 bytes of data:
```

```
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time=10ms TTL=124
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time=10ms TTL=124
```

```
Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

```
C:\>ping 192.168.29.226
```

```
Pinging 192.168.29.226 with 32 bytes of data:
```

```
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time<1ms TTL=125
Reply from 192.168.29.226: bytes=32 time=10ms TTL=125
```

```
Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ping 192.168.29.227
```

```
Pinging 192.168.29.227 with 32 bytes of data:
```

```
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time<1ms TTL=125
Reply from 192.168.29.227: bytes=32 time=10ms TTL=125
Reply from 192.168.29.227: bytes=32 time=10ms TTL=125
```

```
Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

```
C:\>ping 192.168.29.228
```

```
Pinging 192.168.29.228 with 32 bytes of data:
```

```
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=10ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=1ms TTL=125
```

```
Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ftp 192.168.29.227
```

```
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
```

Physical Config Desktop Programming Attributes

Command Prompt

```
Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

```
C:\>ping 192.168.29.228
```

```
Pinging 192.168.29.228 with 32 bytes of data:
```

```
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=10ms TTL=125
Reply from 192.168.29.228: bytes=32 time<1ms TTL=125
Reply from 192.168.29.228: bytes=32 time=1ms TTL=125
```

```
Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

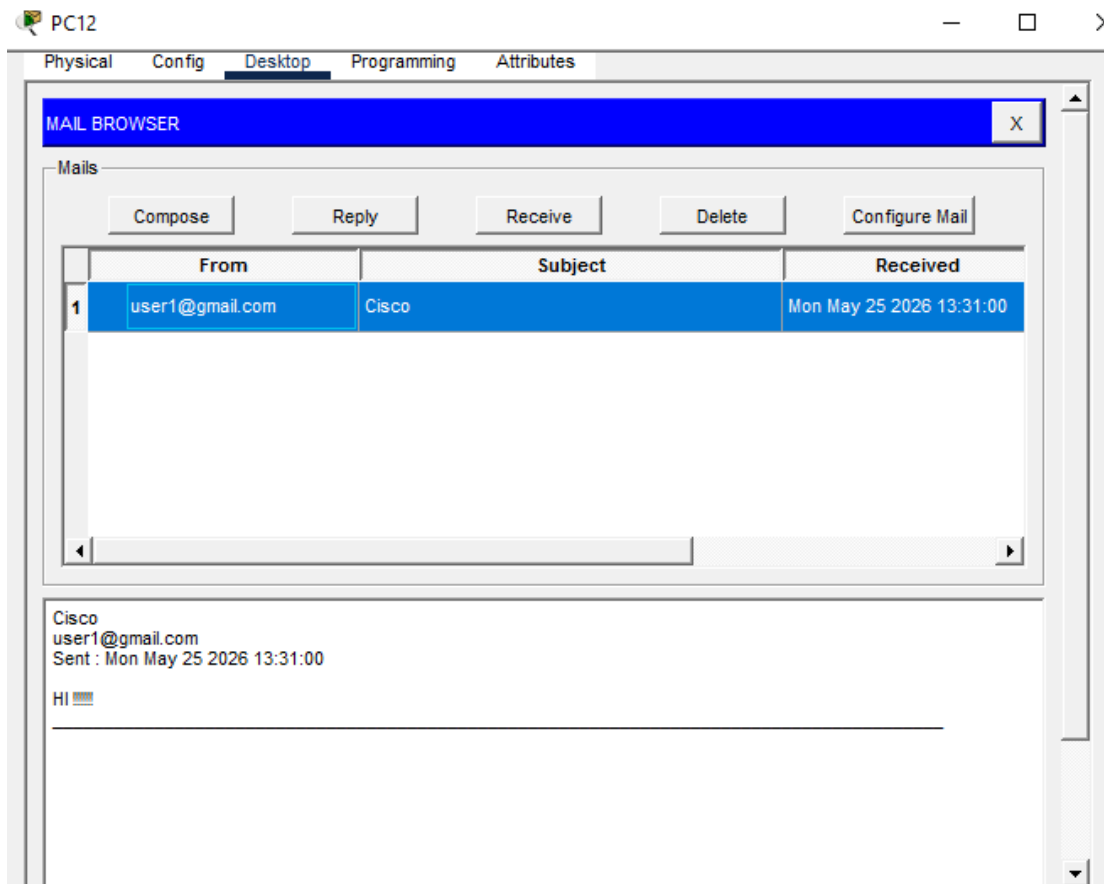
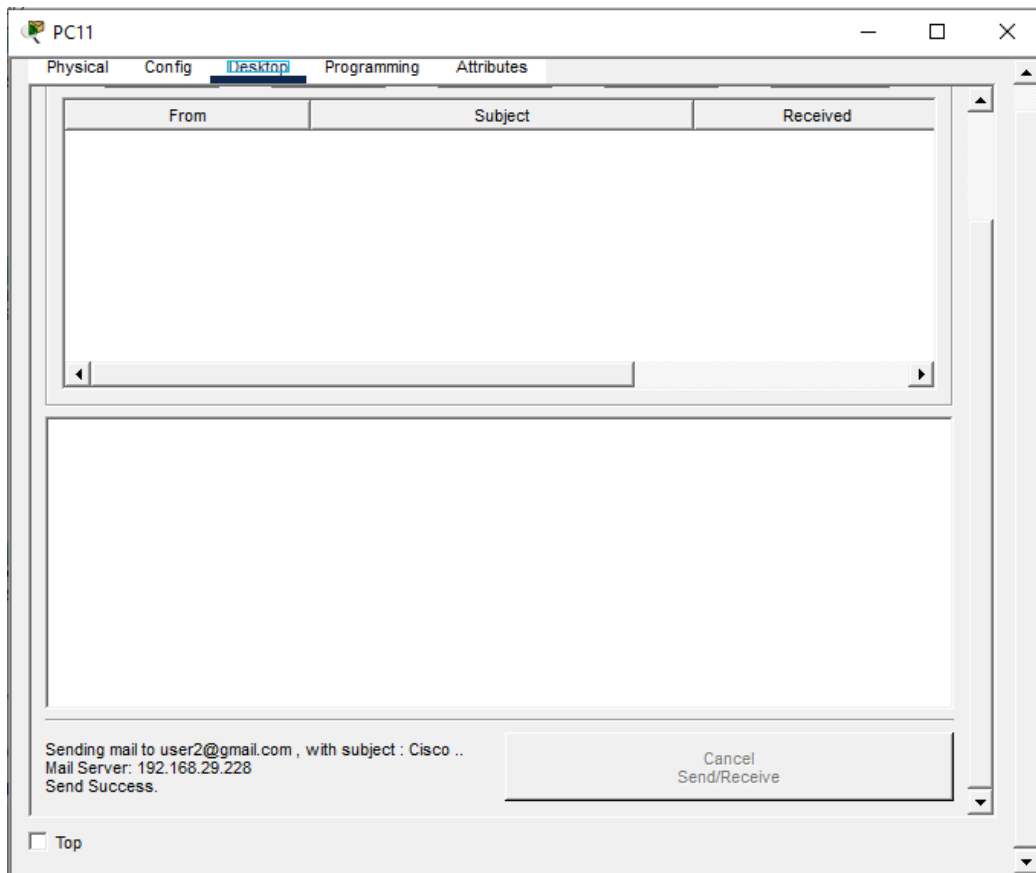
```
C:\>ftp 192.168.29.227
```

```
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir
```

```
Listing /ftp directory from 192.168.29.227:
```

0	: admin.text	28
1	: asa842-k8.bin	5571584
2	: asa923-k8.bin	30468096
3	: c1841-advipservicesk9-mz.124-15.T1.bin	33591768
4	: c1841-ipbase-mz.123-14.T7.bin	13832032
5	: c1841-ipbasek9-mz.124-12.bin	16599160
6	: c1900-universalk9-mz.SPA.155-3.M4a.bin	33591768
7	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
8	: c2600-i-mz.122-28.bin	5571584
9	: c2600-ipbasek9-mz.124-8.bin	13169700
10	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
11	: c2800nm-advipservicesk9-mz.151-4.M4a.bin	33591768
12	: c2800nm-ipbase-mz.123-14.T7.bin	5571584
13	: c2800nm-ipbasek9-mz.124-8.bin	15522644
14	: c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
15	: c2950-i6q412-mz.121-22.EA4.bin	3058048
16	: c2950-i6q412-mz.121-22.EA8.bin	3117390
17	: c2960-lanbase-mz.122-25.FX.bin	4414921
18	: c2960-lanbase-mz.122-25.SEE1.bin	4670455
19	: c2960-lanbasek9-mz.150-2.SE4.bin	4670455
20	: c3560-advipservicesk9-mz.122-37.SEE1.bin	8662192
21	: c3560-advipservicesk9-mz.122-46.SEE1.bin	10713279
22	: c800-universalk9-mz.SPA.152-4.M4a.bin	33591768
23	: c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
24	: cat3k_caa-universalk9.16.03.02.SPA.bin	505532849
25	: cgr1000-universalk9-mz.SPA.154-2.CG	159487552
26	: cgr1000-universalk9-mz.SPA.156-3.CG	184530138
27	: ie9k_iosxe.17.09.04.SPA.bin	596133776
28	: ir800-universalk9-bundle.SPA.156-3.M.bin	160968869
29	: ir800-universalk9-mz.SPA.155-3.M	61750062
30	: ir800-universalk9-mz.SPA.156-3.M	63753767
31	: ir800_yocto-1.7.2.tar	2877440
32	: ir800_yocto-1.7.2.python-2.7.3.tar	6912000
33	: ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg	685645824
34	: pt1000-i-mz.122-28.bin	5571584
35	: pt3000-i6q412-mz.121-22.EA4.bin	3117390

```
ftp>
```



Physical Config Desktop Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=128
Reply from 192.168.29.35: bytes=32 time<1ms TTL=128
Reply from 192.168.29.35: bytes=32 time<1ms TTL=128
Reply from 192.168.29.35: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time=5ms TTL=124
Reply from 192.168.29.227: bytes=32 time=15ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 7ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

```

Physical Config Desktop Programming Attributes

Command Prompt

```

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 7ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=10ms TTL=124

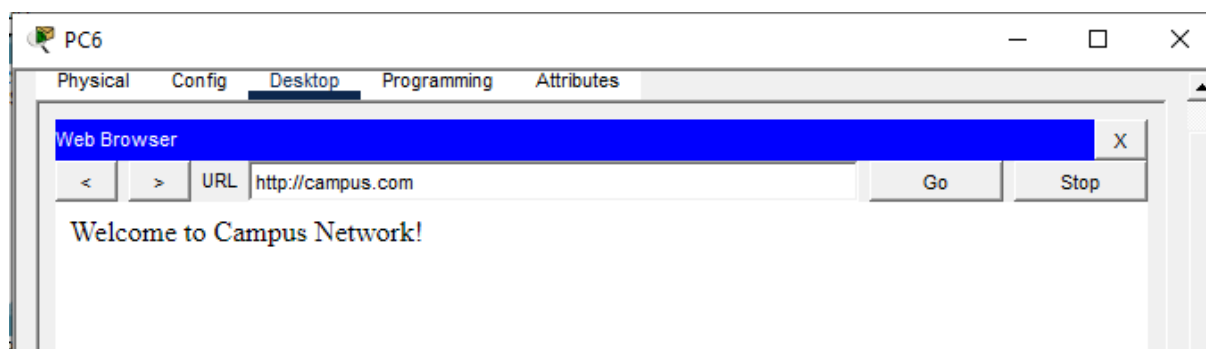
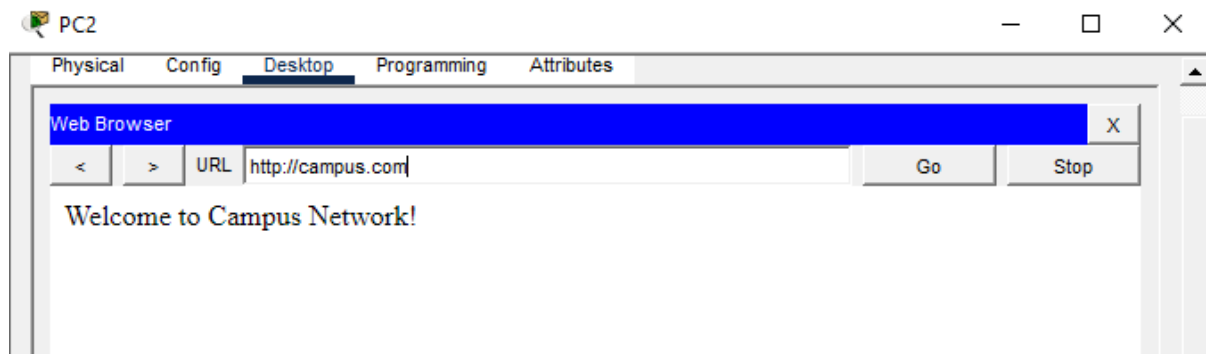
Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username: campus network
331- Username ok, need password
Password: ***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text                               28
 1 : asa842-k8.bin                             5571584
 2 : asa923-k8.bin                             30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin    33591768
 4 : c1841-ipbase-mz.123-14.T7.bin             13832032
 5 : c1841-ipbasek9-mz.124-12.bin              16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin    33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin    33591768
 8 : c2600-i-mz.122-28.bin                     5571584
 9 : c2600-ipbasek9-mz.124-8.bin               13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4a.bin  33591768
12 : c2800nm-ipbase-mz.123-14.T7.bin           5571584
13 : c2800nm-ipbasek9-mz.124-8.bin             15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin    33591768
15 : c2950-i6q412-mz.121-22.EA4.bin           3058048
16 : c2950-i6q412-mz.121-22.EA8.bin           3117390
17 : c2960-lanbase-mz.122-25.FX.bin            4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin          4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin           4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin   8662192
21 : c3560-advipservicesk9-mz.122-46.SE.bin    10713279
22 : c800-universalk9-mz.SPA.152-4.M4a.bin     33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin     83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin    505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG       159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG       184530138
27 : ie9k_iosxe.17.09.04.SPA.bin              596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
29 : ir800-universalk9-mz.SPA.155-3.M          61750062
30 : ir800-universalk9-mz.SPA.156-3.M          63753767
31 : ir800_yocto-1.7.2.tar                     2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar       6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin                    5571584
35 : pt3000-i6q412-mz.121-22.EA4.bin           3117390

ftp>

```



Physical Config Desktop Programming Attributes

Command Prompt

```

28 bytes copied in 0.069 secs (405 bytes/sec)
ftp>
%Error ftp://192.168.29.227/admin.text (Ftp peer reset)

(Disconnecting from ftp server)

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time=1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time=30ms TTL=124

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 30ms, Average = 7ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124
Reply from 192.168.29.227: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.227:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=11ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>

```

Physical Config Desktop Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970    6:0 PM                28          admin.text
1/1/1970    6:0 PM                26          sampleFile.txt
               54 bytes                2 File(s)

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to FT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : asa842-k8.bin                      5571584
 1 : asa923-k8.bin                      30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ipbase-mz.123-14.T7.bin        13832032
 4 : c1841-ipbasek9-mz.124-12.bin        16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin               5571584
 8 : c2600-ipbasek9-mz.124-8.bin         13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin      5571584
12 : c2800nm-ipbasek9-mz.124-8.bin       15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-i6q412-mz.121-22.EA4.bin     3058048
15 : c2950-i6q412-mz.121-22.EA8.bin     3117390
16 : c2960-lanbase-mz.122-25.FX.bin     4414921
17 : c2960-lanbase-mz.122-25.SE1.bin    4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k-caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ie9k_iosxe.17.09.04.SPA.bin        596133776
27 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ir800-universalk9-mz.SPA.155-3.M    61750062
29 : ir800-universalk9-mz.SPA.156-3.M    63753767
30 : ir800_yocto-1.7.2.tar              2877440
31 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ir8340-mono-universalk9_1ot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin             5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin    3117390

ftp>put admin.text

Writing file admin.text to 192.168.29.227:
File transfer in progress...

[Transfer complete - 28 bytes]

28 bytes copied in 0.069 secs (405 bytes/sec)
ftp>
%Error ftp://192.168.29.227/admin.text (Ftp peer reset)

```

Physical Config Desktop Programming Attributes

Command Prompt

```

30 : ir800-universalk9-mz.SPA.156-3.M 6375376
31 : ir800_yocto-1.7.2.tar 2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 6856458
34 : pt1000-i-mz.122-28.bin 5571584
35 : pt3000-i6q412-mz.121-22.EA4.bin 3117390
ftp>get admin.text

```

Reading file admin.text from 192.168.29.227:
File transfer in progress...

[Transfer complete - 28 bytes]

28 bytes copied in 0.01 secs (2800 bytes/sec)

```

ftp>
%Error opening ftp://192.168.29.227/admin.text (Timed out)
.

```

(Disconnecting from ftp server)

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

```

Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127
Reply from 192.168.29.35: bytes=32 time<1ms TTL=127

```

Ping statistics for 192.168.29.35:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

```

Reply from 192.168.29.226: bytes=32 time=1ms TTL=124
Reply from 192.168.29.226: bytes=32 time<1ms TTL=124
Reply from 192.168.29.226: bytes=32 time=10ms TTL=124
Reply from 192.168.29.226: bytes=32 time=1ms TTL=124

```

Ping statistics for 192.168.29.226:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

```

Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time<1ms TTL=124
Reply from 192.168.29.228: bytes=32 time=11ms TTL=124

```

Ping statistics for 192.168.29.228:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password: ***

230- Logged in

(passive mode On)

ftp>dir

Listing /ftp directory from 192.168.29.227:

```

0 : admin.text 28
1 : asa842-k8.bin 5571584
2 : asa923-k8.bin 30468096
3 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
4 : c1841-ipbase-mz.123-14.T7.bin 13832032
5 : c1841-ipbasek9-mz.124-12.bin 16599160
6 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
7 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
8 : c2600-i-mz.122-28.bin 5571584
9 : c2600-ipbasek9-mz.124-8.bin 13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
12 : c2800nm-ipbase-mz.123-14.T7.bin 5571584
13 : c2800nm-ipbasek9-mz.124-8.bin 15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
15 : c2950-i6q412-mz.121-22.EA4.bin 3058048
16 : c2950-i6q412-mz.121-22.EA8.bin 3117390
17 : c2960-lanbase-mz.122-25.FX.bin 4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin 4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
21 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
22 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
27 : ie9k_iosxe.17.09.04.SPA.bin 596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
29 : ir800-universalk9-mz.SPA.155-3.M 61750062
30 : ir800-universalk9-mz.SPA.156-3.M 63753767
31 : ir800_yocto-1.7.2.tar 2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
34 : pt1000-i-mz.122-28.bin 5571584
35 : pt3000-i6q412-mz.121-22.EA4.bin 3117390
ftp>get admin.text

```

Reading file admin.text from 192.168.29.227:

File transfer in progress...

[Transfer complete - 28 bytes]

28 bytes copied in 0.01 secs (2800 bytes/sec)

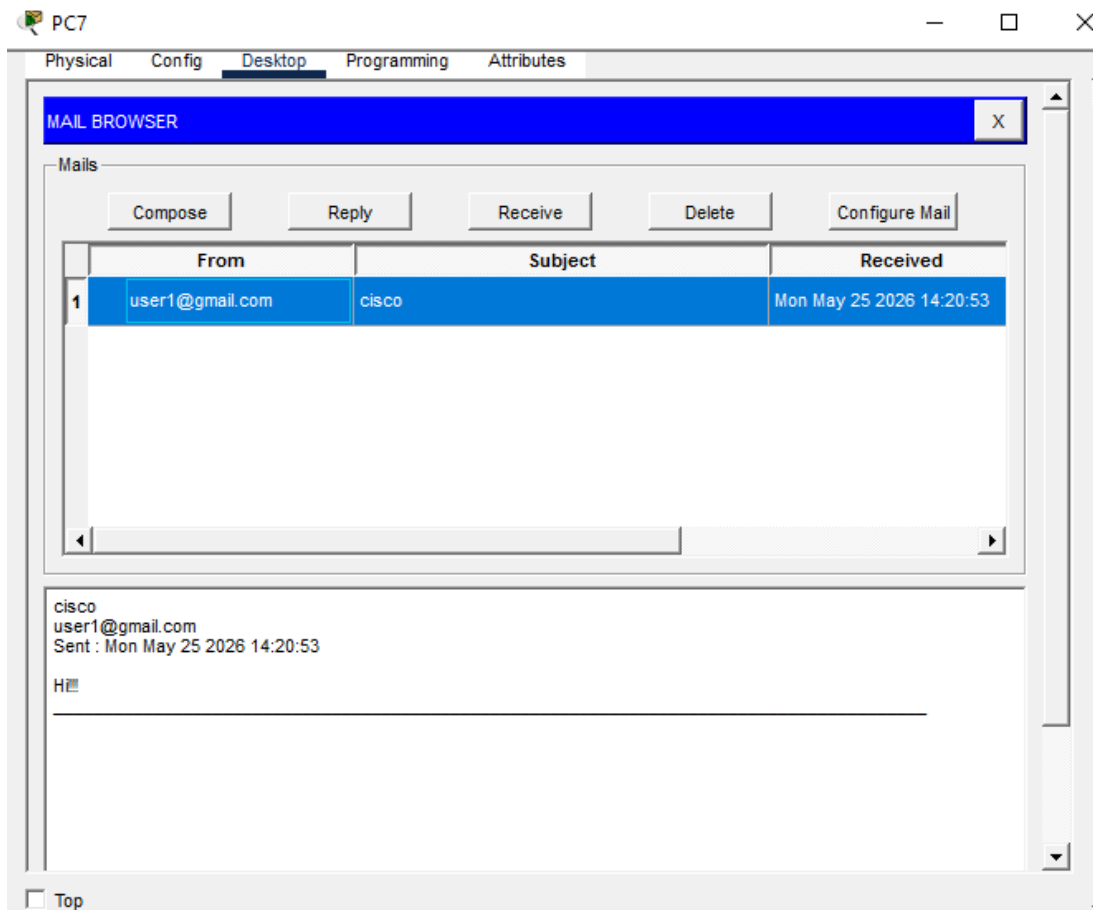
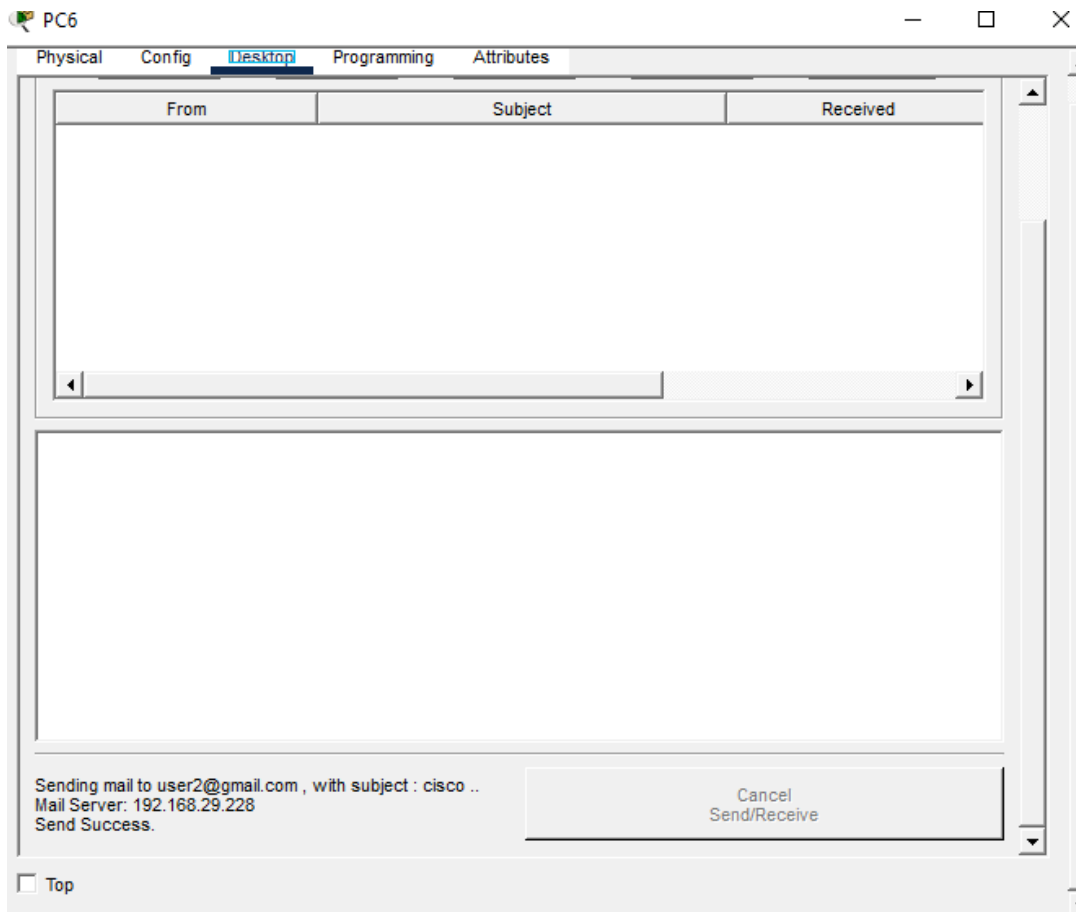
ftp>

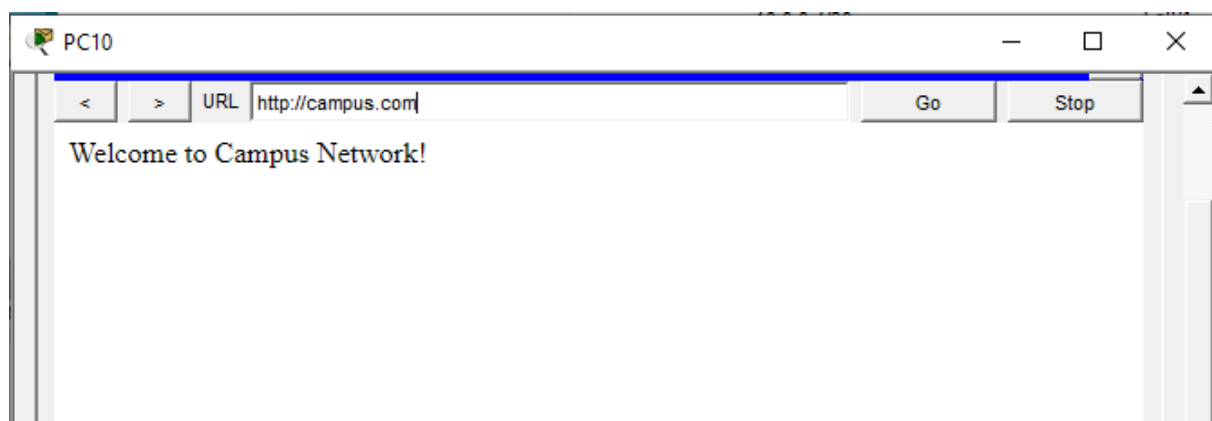
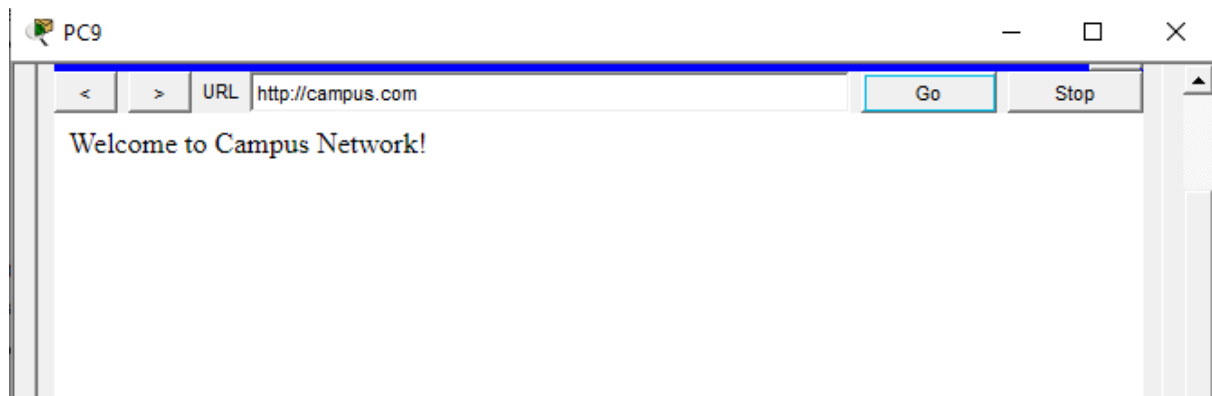
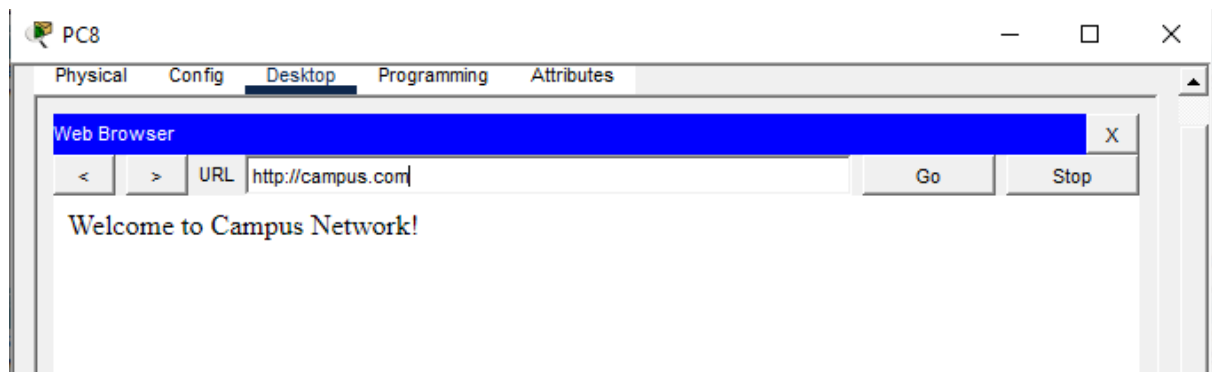
```

%Error opening ftp://192.168.29.227/admin.text (Timed out)
.

```

(Disconnecting from ftp server)





Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
 Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
 Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
 Reply from 192.168.29.35: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.35:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.226: bytes=32 time=15ms TTL=127
 Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.226: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.226:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 192.168.29.227

Pinging 192.168.29.227 with 32 bytes of data:

Request timed out.

Reply from 192.168.29.227: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.227: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.227: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.227:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.228:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password:***

230- Logged in

(passive mode On)

ftp>dir

Command Prompt

Ping statistics for 192.168.29.227:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.228:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password:***

230- Logged in

(passive mode On)

ftp>dir

Listing /ftp directory from 192.168.29.227:

0	: admin.text	28
1	: asa842-k8.bin	5571584
2	: asa923-k8.bin	30468096
3	: c1841-advipservicesk9-mz.124-15.T1.bin	33591768
4	: c1841-ipbase-mz.123-14.T7.bin	13832032
5	: c1841-ipbasek9-mz.124-12.bin	16599160
6	: c1900-universalk9-mz.SPA.155-3.M4a.bin	33591768
7	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
8	: c2600-i-mz.122-28.bin	5571584
9	: c2600-ipbasek9-mz.124-8.bin	13169700
10	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
11	: c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768
12	: c2800nm-ipbase-mz.123-14.T7.bin	5571584
13	: c2800nm-ipbasek9-mz.124-8.bin	15522644
14	: c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
15	: c2950-i6q412-mz.121-22.EA4.bin	3058048
16	: c2950-i6q412-mz.121-22.EA8.bin	3117390
17	: c2960-lanbase-mz.122-25.FX.bin	4414921
18	: c2960-lanbase-mz.122-25.SEE1.bin	4670455
19	: c2960-lanbasek9-mz.150-2.SE4.bin	4670455
20	: c3560-advipservicesk9-mz.122-37.SE1.bin	8662192
21	: c3560-advipservicesk9-mz.122-46.SE.bin	10713279
22	: c800-universalk9-mz.SPA.152-4.M4.bin	33591768
23	: c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
24	: cat3k_caa-universalk9.16.03.02.SPA.bin	505532849
25	: cgr1000-universalk9-mz.SPA.154-2.CG	159487552
26	: cgr1000-universalk9-mz.SPA.156-3.CG	184530138
27	: ie9k_iosxe.17.09.04.SPA.bin	596133776
28	: ir800-universalk9-bundle.SPA.156-3.M.bin	160968869
29	: ir800-universalk9-mz.SPA.155-3.M	61750062
30	: ir800-universalk9-mz.SPA.156-3.M	63753767
31	: ir800_yocto-1.7.2.tar	2877440
32	: ir800_yocto-1.7.2_python-2.7.3.tar	6912000
33	: ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg	685645824
34	: pt1000-i-mz.122-28.bin	5571584
35	: pt3000-i6q412-mz.121-22.EA4.bin	3117390

ftp>

Physical Config Desktop Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
Reply from 192.168.29.35: bytes=32 time=20ms TTL=124
Reply from 192.168.29.35: bytes=32 time=11ms TTL=124

Ping statistics for 192.168.29.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 7ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
Reply from 192.168.29.226: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text
 1 : asa842-k8.bin
 2 : asa923-k8.bin
 3 : c1841-advipservicesk9-mz.124-15.T1.bin
 4 : c1841-ibase-mz.123-14.T7.bin
 5 : c1841-ibasek9-mz.124-12.bin
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin
 7 : c2600-advipservicesk9-mz.124-15.T1.bin
 8 : c2600-i-mz.122-28.bin
 9 : c2600-ibasek9-mz.124-8.bin
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin

```

Physical Config Desktop Programming Attributes

Command Prompt

```

Ping statistics for 192.168.29.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
Reply from 192.168.29.228: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.228:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 192.168.29.227
Trying to connect...192.168.29.227
Connected to 192.168.29.227
220- Welcome to PT Ftp server
Username:campus network
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.29.227:
 0 : admin.text
 1 : asa842-k8.bin
 2 : asa923-k8.bin
 3 : c1841-advipservicesk9-mz.124-15.T1.bin
 4 : c1841-ibase-mz.123-14.T7.bin
 5 : c1841-ibasek9-mz.124-12.bin
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin
 7 : c2600-advipservicesk9-mz.124-15.T1.bin
 8 : c2600-i-mz.122-28.bin
 9 : c2600-ibasek9-mz.124-8.bin
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin
12 : c2800nm-ibase-mz.123-14.T7.bin
13 : c2800nm-ibasek9-mz.124-8.bin
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin
15 : c2950-i6q412-mz.121-22.EA4.bin
16 : c2950-i6q412-mz.121-22.EA8.bin
17 : c2960-lanbase-mz.122-25.FX.bin
18 : c2960-lanbase-mz.122-25.SEE1.bin
19 : c2960-lanbasek9-mz.150-2.SE4.bin
20 : c3560-advipservicesk9-mz.122-37.SE1.bin
21 : c3560-advipservicesk9-mz.122-46.SE1.bin
22 : c800-universalk9-mz.SPA.152-4.M4.bin
23 : c800-universalk9-mz.SPA.154-3.M6a.bin
24 : cat3k_caa-universalk9.16.03.02.SPA.bin
25 : cgr1000-universalk9-mz.SPA.154-2.CG
26 : cgr1000-universalk9-mz.SPA.156-3.CG
27 : ie9k_iosxe.17.09.04.SPA.bin
28 : ir800-universalk9-bundle.SPA.156-3.M.bin
29 : ir800-universalk9-mz.SPA.155-3.M
30 : ir800-universalk9-mz.SPA.156-3.M
31 : ir800_yocto-1.7.2.tar
32 : ir800_yocto-1.7.2_python-2.7.3.tar
33 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg
34 : pt1000-i-mz.122-28.bin
35 : pt3000-i6q412-mz.121-22.EA4.bin

```

5571584

13169700

50938004

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.29.35

Pinging 192.168.29.35 with 32 bytes of data:

Reply from 192.168.29.35: bytes=32 time<1ms TTL=124
 Reply from 192.168.29.35: bytes=32 time=11ms TTL=124
 Reply from 192.168.29.35: bytes=32 time=10ms TTL=124
 Reply from 192.168.29.35: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.29.35:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.29.226

Pinging 192.168.29.226 with 32 bytes of data:

Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.226: bytes=32 time<1ms TTL=127
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Ping statistics for 192.168.29.226:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.29.228

Pinging 192.168.29.228 with 32 bytes of data:

Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time=1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127
 Reply from 192.168.29.228: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.29.228:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password: ***

230- Logged in

(passive mode On)

ftp>dir

Listing /ftp directory from 192.168.29.227:

0	: admin.text	28
1	: asa842-k8.bin	5571584
2	: asa923-k8.bin	30468096
3	: c1841-advipservicesk9-mz.124-15.T1.bin	33591768
4	: c1841-ipbase-mz.123-14.T7.bin	13832032
5	: c1841-ipbasek9-mz.124-12.bin	16599160
6	: c1900-universalk9-mz.SPA.155-3.M4a.bin	33591768
7	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
8	: c2600-i-mz.122-28.bin	5571584
9	: c2600-ipbasek9-mz.124-8.bin	13169700
10	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
11	: c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768

C:\>ftp 192.168.29.227

Trying to connect...192.168.29.227

Connected to 192.168.29.227

220- Welcome to PT Ftp server

Username: campus network

331- Username ok, need password

Password: ***

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ftp>dir

Listing /ftp directory from 192.168.29.227:

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7	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
8	: c2600-i-mz.122-28.bin	5571584
9	: c2600-ipbasek9-mz.124-8.bin	13169700
10	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
11	: c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768
12	: c2800nm-ipbase-mz.123-14.T7.bin	5571584
13	: c2800nm-ipbasek9-mz.124-8.bin	15522644
14	: c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
15	: c2950-i6q412-mz.121-22.EA4.bin	3058048
16	: c2950-i6q412-mz.121-22.EA8.bin	3117390
17	: c2960-lanbase-mz.122-25.FX.bin	4414921
18	: c2960-lanbase-mz.122-25.SEE1.bin	4670455
19	: c2960-lanbasek9-mz.150-2.SE4.bin	4670455
20	: c3560-advipservicesk9-mz.122-37.SE1.bin	8662192
21	: c3560-advipservicesk9-mz.122-46.SE.bin	10713279
22	: c800-universalk9-mz.SPA.152-4.M4.bin	33591768
23	: c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
24	: cat3k_caa-universalk9.16.03.02.SPA.bin	505532849
25	: cgr1000-universalk9-mz.SPA.154-2.CG	159487552
26	: cgr1000-universalk9-mz.SPA.156-3.CG	184530138
27	: ie9k_iosxe.17.09.04.SPA.bin	596133776
28	: ir800-universalk9-bundle.SPA.156-3.M.bin	160968869
29	: ir800-universalk9-mz.SPA.155-3.M	61750062
30	: ir800-universalk9-mz.SPA.156-3.M	63753767
31	: ir800_yocto-1.7.2.tar	2877440
32	: ir800_yocto-1.7.2_python-2.7.3.tar	6912000
33	: ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg	685645824
34	: pt1000-i-mz.122-28.bin	5571584
35	: pt3000-i6q412-mz.121-22.EA4.bin	3117390

ftp>

Discussion:

1. The campus network was designed and implemented successfully using Cisco Packet Tracer. The network includes several departments connected through routers and switches. Each department was assigned a separate subnet to improve network organization and performance.
2. The Administration Department, CSE Department, EEE Department, Business Studies Department, Laboratory, Library, and Server Room were connected to the network. DHCP services were configured in selected departments to provide automatic IP address assignment to users.
3. VLAN technology was implemented to separate teachers and students into different logical groups. This improved network security and reduced unnecessary broadcast traffic.
4. Several routers were connected through point-to-point links using /30 subnet masks. These routers were responsible for forwarding traffic between different networks. Proper routing configurations ensured successful communication among all departments.
5. A DNS server was configured to provide name resolution services. Additionally, HTTP, FTP, and SMTP servers were installed to provide web, file transfer, and email services. These servers simulated a real enterprise network environment.
6. After completing all configurations, connectivity tests were performed. Devices successfully received IP addresses from DHCP servers, DNS services responded correctly, and communication between departments was verified through ping tests. The successful results demonstrated that the network was functioning properly and meeting all project requirements.

Conclusion:

The implementation of this campus network provided valuable practical experience in network design and configuration. Various networking technologies such as subnetting, VLAN, routing, DHCP, DNS, and server services were integrated into a single infrastructure. The project describes that how different departments can communicate efficiently while maintaining proper network organization and security.

Overall, the project helped in understanding how large networks are designed and managed in real-world organizations. It also improved practical knowledge of network configuration.

References:

- 1/ Class Lecture Notes.
- 2/ Pdf from GCR Lecture (2-5).

Submitted by :

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